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Machine Learning and Deep Learning for Wildfire Spread Prediction: A Review

Henintsoa S. Andrianarivony | M. Akhloufi | Environmental Science, Computer Science · Fire · 18 December 2024

TLDR A systematic **review** of recent ML and DL techniques developed for **wildfire spread prediction** is presented, detailing the commonly used datasets, the improvements achieved, and the limitations of current methods. Expand

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A Review of Genetic Algorithm Approaches for Wildfire Spread Prediction Calibration

Jorge Pereira | Jérôme Mendes | J. S. Junior | C. Viegas | J. Paulo

Environmental Science, Computer Science · Mathematics · 19 January 2022

TLDR A literature **review** on optimization methodologies for **wildfire spread prediction** based on the use of evolutionary algorithms for input parameter set calibration of the Rothermel model’s input parameters, namely: surface-area-to-volume ratio, fuel bed depth, fuel moisture, and midflame wind speed is conducted. Expand

39 | [PDF] | PDF | Save | Cite

Methods for Wildfire Spread Prediction and Their Integration With Remote Sensing Data

S. Khvostikov | S. Bartalev | Environmental Science · Information Technologies in Remote Sensing of the… · 2019

This short **review** article presents description of a field of **wildfire** modelling and use of remote sensing in this field. There are many **wildfire** models based on various approaches (physical,… Expand

3 | PDF | Publisher | Save | Cite

Trending and emerging prospects of physics-based and ML-based wildfire spread models: a comprehensive review

Harikesh Singh | Li-Minn Ang | +4 authors | S. Srivastava | Environmental Science, Physics · Journal of Forest Research · 27 September 2024

The significant threat of **wildfires** to forest ecology and biodiversity, particularly in tropical and subtropical regions, underscores the necessity for advanced predictive models amidst shifting… Expand

19 | Publisher | Save | Cite

A Comprehensive Review of Empirical and Dynamic Wildfire Simulators and Machine Learning Techniques used for the Prediction of Wildfire in Australia

Harikesh Singh | Li-Minn Ang | D. Paudyal | Mauricio Acuna | P. K. Srivastava | S. Srivastava

Environmental Science, Computer Science · Technology, Knowledge and Learning · 2 April 2025

TLDR This **review** article provides a comprehensive analysis of various empirical and dynamic **wildfire** simulators alongside machine learning techniques employed for **wildfire prediction** in Australia, examining the effectiveness of traditional empirical methods, dynamic physical models, and advanced ML algorithms in forecasting **wildfire spread** and behaviour. Expand

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Generative AI as a Pillar for Predicting 2D and 3D Wildfire Spread: Beyond Physics-Based Models and Traditional Deep Learning

Haowen Xu Sisi Zlatanova Ruiyu Liang I. Canbulat

Environmental Science, Computer Science · Fire · 3 June 2025

TLDR This paper explores how generative artificial intelligence models—such as GANs, VAEs, and transformers—can serve as transformative tools for **wildfire prediction** and simulation, and advocates for a paradigm shift toward multimodal generative frameworks to support proactive, data-informed **wildfire** response. Expand

2 [PDF] arXiv Save Cite

Prediction of global spread of COVID-19 pandemic: a review and research challenges

Saloni Shah Aos Mulahuwaish K. Ghafoor H. S. Maghdid Medicine, Environmental Science · Artificial Intelligence Review · 19 August 2020

TLDR This attempt to compile different research methods and comparing their accuracy in predicting the **spread** of COVID-19 shows what accuracy of research brings to enable fighting this disease; while on the other hand, it also shows what lack of response from local authorities can do in **spread**ing this virus. Expand

11 PDF Springer Nature Save Cite

AI for Wildfire Management: From Prediction to Detection, Simulation, and Impact Analysis—Bridging Lab Metrics and Real-World Validation

Nicolas Caron Hassan N. Noura Lise Nakache Christophe Guyeux Benjamin Aynes

Environmental Science, Computer Science · Applied Informatics · 1 October 2025

TLDR By bridging the gap between laboratory-oriented models and real-world validation, this work advances prior **review**s and aims to strengthen confidence in AI-driven **wildfire** management while guiding future research toward operational applicability. Expand

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Deep Learning Approaches for Wildland Fires Using Satellite Remote Sensing Data: Detection, Mapping, and Prediction

Rafik Ghali M. Akhloufi Environmental Science, Computer Science · Fire · 7 May 2023

TLDR A comprehensive **review** of recent deep learning techniques for detecting, mapping, and predicting wildland fires using satellite remote sensing data, including fire detection and mapping, severity estimation, and **spread prediction** is provided. Expand

58 [PDF] PDF Save Cite

USING IOT AND ML FOR FOREST FIRE DETECTION, MONITORING, AND PREDICTION: A LITERATURE REVIEW

Mounir Grari Mimoun Yandouzi +4 authors And Mimoun Moussaoui

Environmental Science, Computer Science

TLDR This paper examines literature proposals utilizing IoT and DL to detect **wildfire**s and their **spread** via a comprehensive evaluation and comparison of existing works and focuses on this direction of research. Expand

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